Quiz 1 **ECE 411, Spring 2025**

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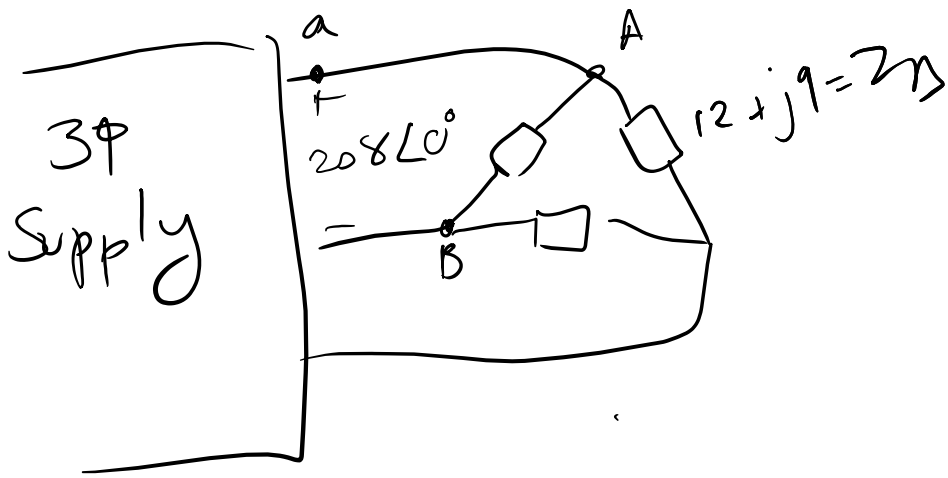
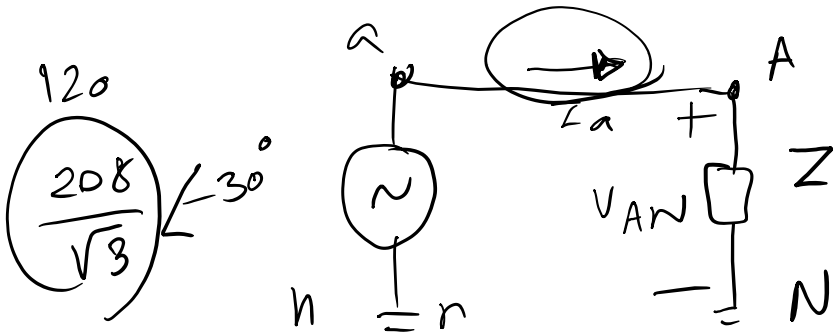
Solution:





$$\frac{208}{\sqrt{3}}\angle -30^\circ \quad 120$$

$$Z_Y = \frac{12+j9}{3} = 4+3j$$

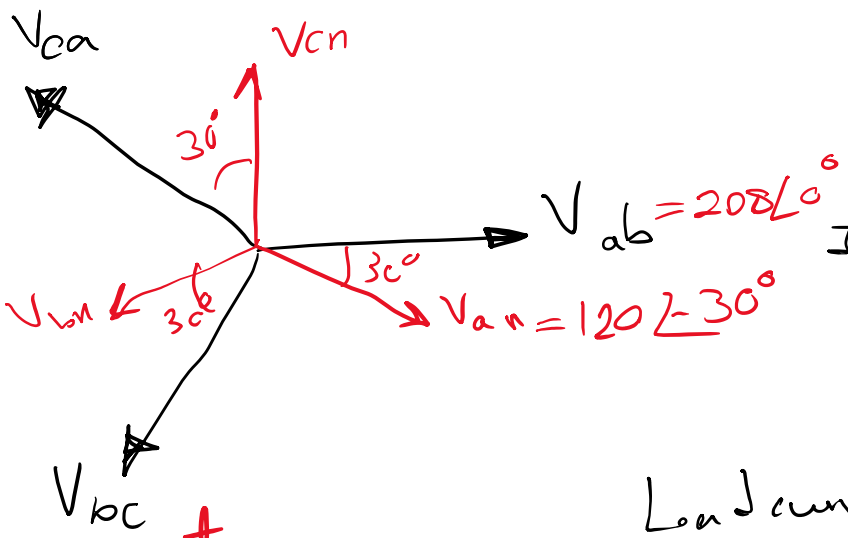
$$I_a = \frac{V_{an}}{Z_{AN}} = \frac{120\angle -30^\circ}{4+3j} = \frac{120\angle -30^\circ}{5\angle 36.87^\circ} = 24\angle -66.8^\circ \ A$$

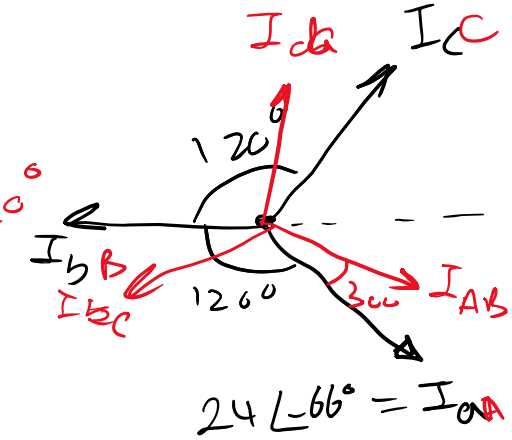
$$S_{1\varphi} = VI^* = (120\angle -30^\circ)(24\angle -66.87^\circ)^* = 2880\angle 36.87^\circ \ VA$$

$$S_{3\varphi} = 3S_{1\varphi} = 3VI^* = 3(2880)\angle 36.87^\circ = 8640\angle 36.87^\circ$$

$$\boxed{S_{3\varphi} = \underline{6912}\ W + j\underline{5184}\ VAR}$$

$$pf = \cos(\underbrace{36.87^\circ}_{\text{lagging}}) = 0.8$$

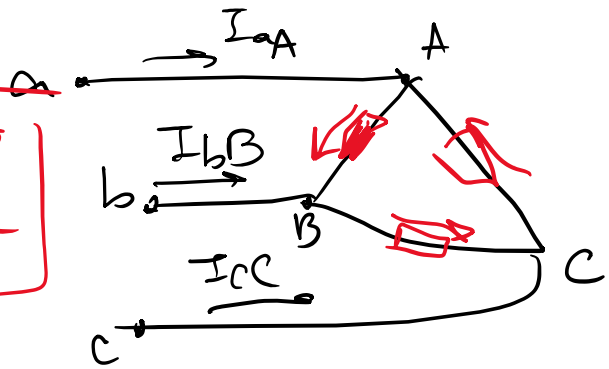




Load currents $\stackrel{?}{=}$ Line Currents

Load Current

$$\boxed{I_{AB} = \frac{24}{\sqrt{3}} \angle -36.87°}$$



$|I_{aA}| = \sqrt{3}\,|I_{AB}|$

$\angle I_{aA} \stackrel{?}{=} \angle I_{AB} - 30°$

$-66.87° = \angle I_{AB} - 30°$

$\angle I_{AB} = -36.87°$

A balanced Δ-connected impedance load with $(12 + j9)\ \Omega$ per phase is supplied by a balanced three-phase 60-Hz, 208-V source. (a) Calculate the line current, the total real and reactive power absorbed by the load, the load power factor, and the apparent load power. (b) Sketch a phasor diagram showing the line currents, the line-to-line source voltages, and the Δ-load currents. Assume positive sequence and use V_{ab} as the reference.